Chapter 16 Notes

Section 1

Theorem 2.7

Here is a proof of the classical part of this theorem. Consider some model modal $\mathcal{M}$ for which the hypotheses of the problem hold

Atomic case: In this case X is atomic. Thus it must be that $A = X$ and $Y = B$, meaning, because we have assumed $A \equiv B$ is valid in $\mathcal{M}$, $X \equiv Y$ (which is the same formula) is also valid in $\mathcal{M}$.

Negation case: In this case X is of the form $\sim\phi$. Then either $A = X$ or A is a subformula of ϕ. In the first case, obviously it follows that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, Y must be of the form $\sim\psi$ (where $\phi \equiv \psi$ is valid over $\mathcal{M}$ by assumption). Then $\phi \supset \psi$ is valid over $\mathcal{M}$ (by assumption), meaning $\sim\psi \supset \sim\phi$ is valid over $\mathcal{M}$. Similarly, $\sim\phi \supset \sim\psi$ is valid over $\mathcal{M}$ (because $\psi \supset \phi$ is valid over $\mathcal{M}$). Thus $\sim\phi \equiv \sim\psi$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Conjunction case: In this case X is of the form $\phi \wedge \psi$. Then either $A = X$ or A is a subformula of either ϕ or ψ. In the first case, obviously it follwos that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, Y must be of the form $\theta \wedge \gamma$, where both $\theta \equiv \phi$ and $\gamma \equiv \psi$ are true in $\mathcal{M}$. Then $(\phi \wedge \psi) \equiv (\theta \wedge \gamma)$, that is, $X \equiv Y$, being valid in $\mathcal{M}$ follows easily.

Existential case: In this case X is of the form $(\exists x)\phi(x)$. Then either $A = X$ or A is a subformula of $\phi(x)$. Again, the first case is obvious. In the second case, we show that (1) $(\exists x)\phi(x) \supset (\exists x)\psi(x)$ is valid over $\mathcal{M}$. So, for any world p of the model, suppose that $(\exists x)\phi(x)$ is true at p. Then $\phi(a)$ is true at p for some $a \in \mathcal{D}$. It is easy to see that this means $\psi(a)$ msut be true at p (because $A \equiv B$ is assumed to be valid over $\mathcal{M}$). Thus $(\exists x)\psi(x)$ is true at p. Hence, because (1) is true at all possible worlds of $\mathcal{M}$, it follows that it is valid over $\mathcal{M}$. Similar reasoning shows that the converse of (1) is valid over $\mathcal{M}$. Hence $(\exists x)\phi(x) \equiv (\exists x)\psi(x)$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Exercise 2.1

(1) $p \vDash (X \vee Y)$ iff $p \vDash \sim(\sim X \wedge \sim Y)$ (by definition of $\vee$) iff $p \nvDash (\sim X \wedge \sim Y)$ iff either $p \nvDash \sim X$ or $p \nvDash \sim Y$ iff either $p \vDash X$ or $p \vDash Y$.

(2) $p \vDash (X \supset Y)$ iff $p \vDash \sim(X \wedge \sim Y)$ (by definition of $\supset$) iff $p \nvDash (X \wedge \sim Y)$ iff either $p \nvDash X$ or $p \nvDash \sim Y$ iff either $p \nvDash X$ or $p \vDash Y$.

(3) $p \vDash (\forall x)\phi(x)$ iff $p \vDash \sim(\exists x)\sim\phi(x)$ (by definition of $\forall$) iff $p \nvDash (\exists x)\sim\phi(x)$ iff $p \nvDash \sim\phi(d)$ for each $d \in \mathcal{D}$ iff $p \nvDash \phi(d)$ for each $d \in \mathcal{D}$.

(4) $p \vDash \Diamond X$ iff $p \vDash \sim\Box\sim X$ (by definition of $\Diamond$) iff $p \nvDash \Box\sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \nvDash \sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$.

Exercise 2.2

(1) We first prove the validity of the K axiom over every modal model. Consider some modal model $\mathcal{M}$, and one of its possible worlds p. We must show that $p \vDash \Box(X \supset Y) \supset (\Box X \supset \Box Y)$. It suffices to show that, if $p \vDash \Box(X \supset Y)$, then it

follows that $p \vDash (\Box X \supset \Box Y)$. So suppose $p \vDash \Box(X \supset Y)$. Then this means that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (X \supset Y)$. It suffices to show that, if $p \vDash \Box X$, then it follows that $p \vDash \Box Y$. So suppose that $p \vDash \Box X$. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$. But also $q \vDash (X \supset Y)$, meaning $q \vDash Y$. Thus $p \vDash \Box Y$. Hence $p \vDash (\Box X \supset \Box Y)$ holds, and consequently, $p \vDash \Box(X \supset Y) \supset (\Box \supset \Box Y)$ also holds, and thus $\Box(X \supset Y) \supset (\Box \supset \Box Y)$ is valid.

(2) We now prove the validity of the Barcan formula over every modal model. Consider some modal model $\mathcal{M}$, and one of its possible worlds p. We must show that $p \vDash (\forall x)\Box\phi(x) \supset \Box(\forall x)\phi(x)$. It suffices to show that, if $p \vDash (\forall x)\Box\phi(x)$, then it follows that $p \vDash \Box(\forall x)\phi(x)$. So suppose that $p \vDash (\forall x)\Box\phi(x)$. Thus $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$, meaning $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. To show that $p \vDash \Box(\forall x)\phi(x)$ holds, we must show that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$. So consider some such q. We must show that $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. But we have already shown this. Thus $q \vDash (\forall x)\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x)$, and consequently, the Barcan formula is valid.

Exercise 2.3

We show that the converse of the Barcan formula, that is, $\Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, is valid over every modal model. Consider any modal model $\mathcal{M}$, and one of its possible worlds p. It suffices to show that, if $p \vDash \Box(\forall x)\phi(x)$ holds, then $p \vDash (\forall x)\Box\phi(x)$ also holds. So suppose that $p \vDash \Box(\forall x)\phi(x)$ holds. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$ holds, meaning $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. To show that $p \vDash (\forall x)\Box\phi(x)$ holds, we must show that $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$. So consider some such d. Then we must show that $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. So consider some such q. But we already know that $q \vDash \phi(d)$ holds. Thus $p \vDash \Box\phi(d)$, meaning $p \vdash (\forall x)\Box\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, and thus the converse of the Barcan formula is valid.

Exercise 2.4

For this, we first prove by induction (using Exercise 2.3) that

$$(\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n) \equiv \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$$

is valid. The base case is just Exercise 2.3. Now suppose the result holds for some n. Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \equiv \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid, meaning we must show that both

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

and

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

are valid.

On the one hand, we show that

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid. So suppose

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

Then, by Exercise 2.3 and the Replacement Theorem,

$$p \vDash (\forall x_1)...(\forall x_n)\Box(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

meaning, by the induction assumption,

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

Hence

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid.

On the other hand, we show that

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

is valid. So suppose that

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

Then

$$p \vDash (\forall x_1)\Box(\forall x_2)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

by Exercise 2.3, and then

$$p \vDash (\forall x_1)...(\forall x_{n+1}\Box\phi(x_1, ..., x_{n+1})$$

by the induction assumption and the Replacement Theorem. Hence

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

is valid.

Thus we know that

$$(\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n) \equiv \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$$

is valid. Then $p \vDash \Box\phi(x_1, ..., x_n)$ iff $p \vDash (\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n)$ iff $p \vDash \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$ iff $q \vDash (\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$ iff $q \vDash \phi(x_1, ..., x_n)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$.

Exercise 2.5

(a) We show that $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ is valid over every modal model. So consider some modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash \Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$. It suffices to show that, if $p \vDash \Box(X \wedge Y)$, then $p \vDash (\Box X \wedge \Box Y)$. So suppose that $p \vDash \Box(X \wedge Y)$. Thus $q \vDash (X \wedge Y)$ holds for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. This means that $q \vDash X$ and $q \vDash Y$ both hold for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, hence both $p \vDash \Box X$ and $p \vDash \Box Y$ hold, and consequently, $p \vDash (\Box X \wedge \Box Y)$ holds. Thus $p \vDash \Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$, meaning $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ is valid over $\mathcal{M}$, hence valid over every modal model.

(b) We show that $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ is valid over every modal model. So consider some modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash (\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$. It suffices to show that, if $p \vDash (\Box X \wedge \Box Y)$, then $p \vDash \Box(X \wedge Y)$. So suppose that $p \vDash (\Box X \wedge \Box Y)$. Then $p \vDash \Box X$ and $p \vDash \Box Y$. Hence $q \vDash X$ and $q \vDash Y$ both hold for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. This means $q \vDash (X \wedge Y)$ holds for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. But this means $p \vDash \Box(X \wedge Y)$ holds, and thus $p \vDash (\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$, meaning $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ is valid over $\mathcal{M}$, hence valid over every modal model.

(c) We show that $(\Diamond \vee \Diamond Y) \equiv \Diamond(X \vee Y)$ is valid over every modal model. Consider any modal model $\mathcal{M}$, and any of its possible worlds p.

On the one hand, we show that $p \vDash (\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$. It suffices to show that, if $p \vDash (\Diamond \vee \Diamond Y)$, then $p \vDash \Diamond(X \vee Y)$. So suppose that $p \vDash (\Diamond X \vee \Diamond Y)$. Thus either $p \vDash \Diamond X$ or $p \vDash \Diamond Y$, hence (1) there is either some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash X$, or there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash Y$. To show that $p \vDash \Diamond(X \vee Y)$ holds, we must show that there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash (X \vee Y)$. But, we are guaranteed to have such a q by (1). Hence $p \vDash \Diamond(X \vee Y)$ holds, meaning $p \vDash (\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$, and thus $(\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$ is valid over all modal models.

On the other hand, we show that $p \vDash \Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$. It suffices to show that, if $p \vDash \Diamond(X \vee Y)$, then $p \vDash \Diamond(X \vee Y)$. So suppose that $p \vDash \Diamond(X \vee Y)$. Then, for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (X \vee Y)$, meaning either $q \vDash X$ or $q \vDash Y$. But this means that there is either some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash X$, or there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash Y$, meaning either $p \vDash \Diamond X$ or $p \vDash \Diamond Y$, and thus $p \vDash (\Diamond X \vee \Diamond Y)$. Hence $p \vDash \Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$, meaning $\Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$ is valid over all modal models.

(d) We first show that $(\Diamond X \wedge \Diamond Y) \supset \Diamond(X \wedge Y)$ is not valid. Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that it could be the case that $p \nvDash (\Diamond X \wedge \Diamond Y) \supset \Diamond(X \wedge Y)$, which is the same as saying that it could be the case that $p \vDash (\Diamond X \wedge \Diamond Y) \wedge \sim\Diamond(X \wedge Y)$. Well, it certainly could be the case that $p \vDash (\Diamond X \wedge \Diamond Y)$. If it is, then either $p \vDash \Diamond X$ or $p \vDash \Diamond Y$, meaning either $q \vDash X$ for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, or $q \vDash Y$ for some $q \in \mathcal{G}$ with $p\mathcal{R}q$. But this does not mean that there must be some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that both $q \vDash X$ and $q \vDash Y$ hold, which is what we would need to know that $p \vDash \Diamond(X \wedge Y)$. Hence we cannot necessarily conclude $p \vDash \Diamond(X \wedge Y)$, meaning $(\Diamond X \wedge \Diamond Y) \supset \Diamond(X \wedge Y)$ is not valid.

Second, we show that $\Box(X \vee Y) \supset (\Box X \vee \Box Y)$ is not valid. Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that it could be the case that $p \nvDash \Box(X \vee Y) \supset (\Box X \vee \Box Y)$, which is the same as saying that it could be the case that $p \vDash \Box(X \vee Y) \wedge \sim(\Box X \vee \Box Y)$. Well, it certainly could be the case that $\Box(X \vee Y)$. If it is, then $q \vDash (X \vee Y)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, meaning for each such q, either $q \vDash X$ or $q \vDash Y$. But this does not mean that either $q \vDash X$ holds for all such q, or $q \vDash Y$ holds for all such q, which is what we would need for $p \vDash (\Box X \vee \Box Y)$. Hence we cannot necessarily conclude $p \vDash (\Box X \vee \Box Y)$, meaning $(\Box X \vee \Box Y)$ is not valid.

Exercise 2.6

We show that $\Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$ is valid in every modal model (note that the converse is not valid). Consider any modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash \Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$. It suffices to show that, if $p \vDash \Box(X \equiv Y)$, then $p \vDash (\Box X \equiv \Box Y)$. So suppose that $p \vDash \Box(X \equiv Y)$. Thus (1) $q \vDash (X \equiv Y)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. To show that $p \vDash (\Box X \equiv \Box Y)$, we must show that $p \vDash (\Box X \supset \Box Y)$, and $p \vDash (\Box Y \supset \Box X)$.

On the one hand, suppose that $p \vDash \Box X$. Thus $q \vDash X$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, but this also means (by (1)) that $q \vDash Y$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, and thus $p \vDash \Box Y$.

The case for showing $p \vDash (\Box Y \supset \Box X)$ is similar to the first case.

Thus we have both $p \vDash (\Box X \supset \Box Y)$ and $p \vDash (\Box Y \supset \Box X)$, meaning $p \vDash (\Box X \equiv \Box Y)$, which is what we were trying to prove.

Exercise 2.7

For this exercise, "Classical logic" as a reason in the derivation stands for something derived due to the assumption that the axiom system we are working in contains the standard rules for classical logic.

(a)

1.	$((X \wedge Y) \supset X)$	(Classical logic)
2.	$\Box((X \wedge Y) \supset X)$	(1, Necessitation rule)
3.	$\Box(X \wedge Y) \supset \Box X$	(2, K axiom)

(b) For this, note that part (a) can of course be altered to prove that $\Box(X \wedge Y) \supset \Box Y$.

1.	$\Box(X \wedge Y) \supset \Box X$	(Part (a))
2.	$\Box(X \wedge Y) \supset \Box Y$	(Part (a))
3.	$\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$	(1, 2, Classical logic)

(c)

1.	$(X \supset (Y \supset (X \wedge Y)))$	(Classical logic)
2.	$\Box(X \supset (Y \supset (X \wedge Y)))$	(1, Necessitation rule)
3.	$\Box X \supset \Box(Y \supset (X \wedge Y))$	(2, K axiom)
4.	$\Box X \supset (\Box Y \supset \Box(X \wedge Y))$	(3, K axiom)
5.	$(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$	(4, Classical logic)